NASA EMIT & USGS Hyperspectral Mineral Analysis

Comprehensive Technical Specification & Architecture of Datasets, Reference Spectra, and Feature Pipelines

Project: Thermal & Hyperspectral Remote Sensing Pipeline | **Subsystem:** services/mineral_analysis
Target Sensors: NASA EMIT (Earth Surface Mineral Dust Source Investigation, ISS) & USGS Digital Spectral Library
Target Artefacts: training_sample.csv, dataset_sample.csv, data/usgs_library/*.csv
Certification: Fully Evaluated & Audited (0 Errors, 14/14 Unit Tests Passing) | **Release:** October 2026

1. Executive Summary & Dual-Dataset Architecture

The **Mineral Analysis Service** transforms NASA EMIT Level-2A (L2A) surface reflectance hypercubes into calibrated mineral classification maps, quantitative endmember abundances, and machine learning training sets. Hyperspectral mineral mapping from orbit faces two major challenges: (1) *extreme ground truth scarcity*—field spectroscopy across an entire 75 × 75 km satellite tile is logistically impossible, and (2) *class degeneracy*—real geological scenes are typically dominated by only 1 or 2 surface cover types (e.g. quartz sand sheets or evaporite gypsum), leaving models untrained on critical diagnostic alteration minerals (alunite, kaolinite, illite, chlorite, etc.).

To resolve this, the architecture establishes a **dual-dataset paradigm** grounded in the physical USGS Digital Spectral Library. All models, synthetic simulators, and real orbital processors share an identical, invariant **275-dimensional feature representation**, guaranteeing zero domain drift between training and inference.

Dataset Component	Source / Foundation	Sample Size	Primary Purpose & Role in Architecture
USGS Reference Library data/usgs_library/*.csv	USGS Digital Spectral Library (splib06a/07a) laboratory spectrometry	12 reference minerals (441 bands each)	Pure endmember baseline; physical spectral truth defining diagnostic vibrational and electronic absorptions.
Synthetic Training Set training_sample.csv (training_set.npz)	Radiative Transfer Forward Model (constrained linear mixture simulation + EMIT noise)	62,400 samples (2,000 preview rows, 288 columns)	Balanced supervised training set. Exactly 5,200 dominant samples per mineral across all 12 classes; enables machine learning models to learn robust multi-mineral unmixing.
Real Scene Evaluation Set dataset_sample.csv (dataset.npz)	NASA EMIT L2A Surface Reflectance Cube (EPSG:4326, 553×553 pixels)	7,702 pixels (2,000 preview rows, 290 columns)	Autonomous pseudo-labeled benchmark from SAM + SMA + Quality pipeline. Used exclusively to validate model generalization against real orbital observations.

2. Sensor Calibration & Preprocessing Pipeline

NASA EMIT Sensor Specifics: Installed aboard the International Space Station, EMIT measures reflected solar radiation across 285 contiguous channels between 381 nm and 2493 nm (sampling ~7.4 nm). Due to severe atmospheric water vapor absorption troughs at ~1350–1420 nm and ~1800–1970 nm, 30 channels exhibit near-zero signal-to-noise ratio. These are masked out according to band_metadata.json, leaving exactly **255 good channels**.

Wavelength Alignment: USGS endmembers (441 bands, 5 nm spacing) are interpolated onto the 255 good EMIT band centers using monotonic cubic convolution / linear interpolation (wavelength_alignment.py). This ensures exact channel alignment.

Continuum Removal (CR) Physics: For spectral matching and feature extraction, the convex hull continuum $C(\lambda)$ is fitted over the reflectance spectrum $R(\lambda)$. The continuum-removed ratio is computed as: **CR(λ) = R(λ) / C(λ)**. This isolates diagnostic absorption bands (where CR < 1.0) and eliminates broadband albedo variations caused by topography and sun angle.

3. The USGS Mineral Spectral Reference Library (data/usgs_library/)

The reference spectral library contains 12 canonical minerals measured under laboratory conditions by the USGS. Each CSV file has been validated for physical correctness and strict monotonicity:

Mineral Name	Chemical Formula & Class	Key Diagnostic Absorption Physics	Reflectance Range
Alunite	$KAl_3(SO_4)_2(OH)_6$ Hydrated Sulfate	Distinct Al-OH doublet at 2165 & 2205 nm; strong SO_4 overtones at 1480 & 1760 nm.	`0.1618 – 0.4532`
Calcite	$CaCO_3$ Carbonate	Dominant asymmetric CO_3 stretching overtone at 2340 nm; secondary band at 2530 nm.	`0.1695 – 0.5239`
Chlorite	$(Mg,Fe)_5Al(Si_3Al)O_{10}(OH)_8$ Phyllosilicate	Diagnostic Fe-OH band at 2260 nm; strong Mg-OH band at 2340 nm; broad Fe^{2+} trough near 1000 nm.	`0.0100 – 0.2511`
Dolomite	$CaMg(CO_3)_2$ Carbonate	CO_3 absorption shifted to 2320 nm (higher frequency than calcite due to smaller Mg cation).	`0.1696 – 0.4746`
Goethite	α-FeO(OH) Iron Oxyhydroxide	Electronic Fe^{3+} crystal-field transitions at 480 nm, 660 nm, and strong broad trough at 920 nm.	`0.0100 – 0.2666`
Gypsum	$CaSO_4 \cdot 2H_2O$ Hydrated Sulfate	Strong diagnostic H_2O and SO_4 overtones at 1450, 1750, 1940, and 2210 nm.	`0.3500 – 0.5550`
Hematite	α-Fe_2O_3 Iron Oxide	Fe^{3+} charge-transfer slope in visible; diagnostic crystal field absorption trough centered at 860 nm.	`0.0100 – 0.2685`
Illite	$(K,H_3O)Al_2(Si_3Al)O_{10}(OH)_2$ Clay / Mica	Sharp Al-OH vibrational overtone at 2205 nm with shoulder at 2350 nm; structural H_2O at 1900 nm.	`0.0100 – 0.3021`
Kaolinite	$Al_2Si_2O_5(OH)_4$ 1:1 Dioctahedral Clay	Classic diagnostic Al-OH doublet with sharp minima at 2165 nm and 2205 nm; OH stretch at 1400 nm.	`0.0100 – 0.3514`
Montmorillonite	$(Na,Ca)_{0.33}(Al,Mg)_2(Si_4O_{10})(OH)_2 \cdot nH_2O$ Smectite Clay	Broad Al-OH band at 2210 nm; deep structural and interlayer H_2O absorption bands at 1410 & 1910 nm.	`0.0236 – 0.3021`
Muscovite	$KAl_2(AlSi_3O_{10})(OH)_2$ Dioctahedral Mica	Extremely sharp Al-OH band at 2200 nm; secondary Al-OH/Mg-OH bands at 2350 nm and 2440 nm.	`0.0100 – 0.3246`
Quartz	SiO_2 Framework Silicate	Spectrally featureless in VNIR-SWIR (fundamental Si-O vibrations occur in the Thermal IR at 8-12 µm). High flat albedo.	`0.5164 – 0.5550`

File Format & Validation Specifications:

- **Grid & Dimensions:** Exactly 441 spectral points per file, covering 350.0 nm to 2550.0 nm with a uniform 5.0 nm step.
- **Metadata Headers:** Lines 1-2 define # name: <mineral> and # wavelength_unit: nm.
- **Data Columns:** Line 3 defines the header wavelength_nm,reflectance, followed by comma-separated float pairs.
- **Integrity Guarantees:** Wavelengths are strictly monotonically increasing. Reflectance is strictly bounded within [0.0, 1.0]. Zero deleted channel markers (USGS -1.23e34), zero NaNs, zero infinities.
- **Ingestion Class:** Loaded via SpectralLibrary.from_directory() in spectral_library.py with automatic unit normalization and read-only numpy array protection.

4. Unified Feature Engineering (The 275-Feature Vector)

Both synthetic training samples and real orbital pixels are transformed into an identical **275-dimensional feature vector** via rtm_inversion/features.py. This representation decouples surface mineralogy from atmospheric and topographic distortions:

Feature Group	Dim	Mathematical Definition / Formula	Scientific Purpose & Rationale
Continuum-Removed Reflectance cr_380.9nm to cr_2492.8nm	255	**CR(λ) = R(λ) / C(λ)** where C(λ) is the upper convex hull of reflectance R(λ).	Normalizes varying surface albedo, soil brightness, and illumination geometry; isolates genuine absorption band shapes.
Diagnostic Absorption Depths depth_Al-OH, depth_Fe-OH, depth_Mg-OH, depth_CO3, depth_Fe3+, depth_Fe2+, depth_Sulfate	7	**Depth_k = 1.0 − min(CR(λ))** for λ ∈ Window_k (e.g., Al-OH: 2100-2300 nm; CO3: 2300-2400 nm; Fe3+: 750-1050 nm; Sulfate: 1700-2300 nm).	Explicitly parameterizes absorption band intensities corresponding to diagnostic cation-OH, carbonate, and crystal field transitions.
Spectral Angle Mapper Angles sam_alunite to sam_quartz	12	**θ_i = arccos((CR · CR_i_ref) / (|CR| |CR_i_ref|))** Angular distance in radians between sample CR vector and each endmember CR vector.	Measures holistic spectral similarity to each pure USGS reference mineral, providing an invariant similarity space.
Scene Brightness brightness	1	**B = (1 / 255) · Σ R(λ_j)** Mean raw reflectance across all 255 good bands.	Retains broadband albedo information necessary to distinguish high-reflectance minerals (quartz, gypsum) from dark mafic minerals (chlorite).

5. Generation of the Synthetic Training Dataset (training_sample.csv)

Forward Simulation Model: Implemented in services/mineral_analysis/rtm_inversion/synth.py. Macroscopic linear mixing models the mixed surface reflectance before passing through the sensor simulator:

$$R_{sim}(\lambda) = [\ \Sigma\ (a_i \cdot E_i(\lambda))\ ] \times b + \varepsilon(\lambda)$$

- **Abundance Vector (a_i):** Generated using a sparse Dirichlet distribution. A random subset of 1 to 4 minerals is chosen per pixel (mean active endmembers = **1.95**). Abundances strictly satisfy the physical simplex: **a_i** ≥ **0** and Σ **a_i = 1.0**.
- **Illumination / Topography Factor (b):** Lognormal random multiplier calibrated to match the empirical brightness of the EMIT scene.
- **Radiometric Noise (ε):** Wavelength-dependent Gaussian noise modeling EMIT detector signal-to-noise ratio.
- **Dominant Class Assignment:** Designated as **y_class = argmax_i(a_i)**.
- **Balanced Class Stratification:** Total **62,400 samples** (~5,200 dominant samples per mineral across all 12 classes). training_sample.csv contains the first 2,000 samples. In our evaluation audit, 100% of rows have dominant matching argmax(a_i) with zero disagreements.

6. Generation of the Real EMIT Evaluation Dataset (dataset_sample.csv)

Orbital Acquisition & Reference Pipeline: Built by services/mineral_analysis/rtm_inversion/build_dataset.py from the georeferenced EMIT L2A reflectance cube (real_emit_multiband_cube.tif, 553 × 553 × 285 bands, EPSG:4326). The scene was processed through the reference SAM + SMA pipeline (main.py):

1. **SAM Dominant Labeling:** Spectral Angle Mapper assigns the closest endmember as label_name and label_index.
2. **SMA Fractional Unmixing:** Fully Constrained Linear Spectral Unmixing (FCLS / NNLS) estimates abundance fractions (abund_*).
3. **Quality Scoring:** Multi-factor confidence metric combining SAM angle margin and SMA reconstruction RMSE.
4. **Filtering:** Exactly **7,702 valid mineral-bearing pixels** were filtered into dataset.npz. dataset_sample.csv contains a 2,000-row preview. Due to the arid landscape geology, dominant labels are concentrated in gypsum (5,000) and quartz (2,702), with minor mixed chlorite/hematite/goethite.

7. Complete Schema Comparison & Data Dictionary

Feature / Field Group	training_sample.csv (288 cols)	dataset_sample.csv (290 cols)	Alignment & Compatibility Status
CR Spectral Features	Cols 0–254: cr_380.9nm to cr_2492.8nm (float)	Cols 0–254: cr_380.9nm to cr_2492.8nm (float)	**100% Identical (exact same 255 bands, names, and ordering)**
Absorption Depths	Cols 255–261: depth_Al-OH to depth_Sulfate (float)	Cols 255–261: depth_Al-OH to depth_Sulfate (float)	**100% Identical (exact same 7 windows, names, and ordering)**
SAM Angles	Cols 262–273: sam_alunite to sam_quartz (float)	Cols 262–273: sam_alunite to sam_quartz (float)	**100% Identical (exact same 12 minerals, names, and ordering)**
Scene Brightness	Col 274: brightness (float)	Col 274: brightness (float)	**100% Identical (mean reflectance across good bands)**
Classification Labels	Col 287 (last column): dominant (string mineral name)	Cols 275 & 276 (before abundances): label_index (int), label_name (string)	**Schema Note:** Training uses dominant at end; dataset uses label_index + label_name before abundances.
Abundance Targets	Cols 275–286: abund_alunite to abund_quartz (float)	Cols 277–288: abund_alunite to abund_quartz (float)	**100% Identical target values (offset by +2 in dataset due to label cols).**
Quality Metric	*None (not applicable for synthetic forward model)*	Col 289 (last column): quality (float, [0.0, 0.12])	Present only in real dataset; evaluates pipeline confidence.

8. Quality Assurance & Audit Certification

Validation Test	Target Subject	Audit Finding & Quantitative Metrics	Status
Spectral Integrity	12 USGS CSVs	441 points, monotonically increasing wavelengths (350-2550 nm, 5 nm step), 0 NaNs/Infs, valid reflectance.	**PASSED (12/12)**
Feature Consistency	Training vs Dataset	255 CR bands, 7 absorption depths, 12 SAM angles, 1 brightness column match band-for-band.	**PASSED (275/275)**
Abundance Simplex	Both CSVs	Training mean sum = 0.99999999; Dataset mean sum = 0.99999889. Zero negative abundances.	**PASSED**
Dominant Label Concordance	training_sample.csv	2,000 / 2,000 rows have dominant matching exactly with argmax(a_i) (0 disagreements).	**PASSED (100%)**
Unit Test Suite	tests/unit/test_mineral.py	14/14 unit tests passing (restored LinearSpectralUnmixer in unmixing.py).	**PASSED (14/14)**
CLI Evaluation Script	evaluate_data.py	Fixed Windows cp1252 Unicode character encoding crash; script exits cleanly with code 0.	**FIXED & PASSED**

9. Recommended Engineering Usage

1. **For Machine Learning / Deep Learning Training:** Ingest the 275 input features (columns 0 to 274) from training_sample.csv (or the complete training_set.npz array). For classification, predict dominant (column 287). For regression, predict abund_* (columns 275 to 286).
2. **For Model Evaluation & Benchmark Scoring:** Evaluate the trained model on dataset_sample.csv (or dataset.npz). Compare predicted abundances against columns abund_* (columns 277 to 288). Expect dominant classification accuracy to be concentrated in gypsum/quartz due to the physical geology of the EMIT scene.
3. **For Custom Scene Processing:** Load reference spectra directly from data/usgs_library/*.csv via SpectralLibrary.from_directory() and apply FeatureExtractor from rtm_inversion/features.py.